MATH 101 Unit 1: Expressions, Equations, and Inequalities - Practice 3

Version 202610122020.36 | Name: _____

Complete and show all work on a separate sheet of paper. Record final answers on the answer sheet. Give exact answers and simplify completely. Only a basic four-function calculator may be used. Suggested timed practice: 30 minutes.

Answer key follows the answer sheet. Attempt the test before checking it.

1. Simplify. In (a), x and y are nonzero; give positive exponents.

(a) $\dfrac{6x^{-4}y^{3}}{3xy^{-1}}$

(b) $\sqrt{125}$

2. (a) Expand and simplify. (b) Factor completely.

(a) $(4x-3)(x+4)$

(b) $2x^{2}-7x-15$

3. Solve the inequality and write the solution in interval notation.

$|2x-1|\le 5$

4. Simplify and state all values excluded by the original expression.

$$\frac{x^{2}-16}{x^{2}+2x-24}$$

5. Solve both equations. In (b), state original restrictions and check your answer.

(a) $x^{2}+2x-15=0$

(b) $\frac{6}{2x-8}+\frac{1}{x-4}=4$

Student answer sheet

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Question 1

(a)

10 pts

(b)

10 pts

Question 2

(a)

10 pts

(b)

10 pts

Student answer sheet

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Question 3

Derived bounds	12 pts
Interval	8 pts

Question 4

Expression	14 pts
Excluded values	6 pts

Question 5

(a)	10 pts
(b), with check	10 pts

Answer key

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Question 1

(a) $\frac{2y^4}{x^5}$ 10 pts

(b) $5\sqrt{5}$ 10 pts

Question 2

(a) $4x^2 + 13x - 12$ 10 pts

(b) $(x - 5)(2x + 3)$ 10 pts

Answer key

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Question 3

Derived bounds	$-2 \le x \le 3$	12 pts
Interval	$[-2, 3]$	8 pts

Question 4

Expression	$\frac{x+4}{x+6}$	14 pts
Excluded values	$\{-6, 4\}$	6 pts

Question 5

(a)	$x = -5,\ 3$	10 pts
(b), with check	$x = 5;\ x \neq 4$	10 pts